

Diabetes & pregnancy

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Diabetes & pregnancy

Definition

Gestational diabetes:

Glucose intolerance of variable severity with onset or 1st recognition during pregnancy

Pre-existing diabetes :

Patients with blood sugar abnormalities preceding pregnancy, the vast majority of these will have type I (insulin dependent diabetes), the remainder will have type II (non-insulin dependent) diabetes.

GDM

Gestational diabetes

Risk factors for GDM:



High blood pressure.



Heart disease.



Family history of Type 2 diabetes.



Personal history of GD (in previous pregnancies).



Having obesity or overweight before pregnancy.



Advanced maternal age.



Prediabetes.



Polycystic ovary syndrome (PCOS).

There are usually no obvious warning signs of GDM. If you do have symptoms, they may include:



Frequent urination.



Nausea.



Excessive thirst.



Tiredness.

Cleveland Clinic

Incidence of abnormal carbohydrate metabolism

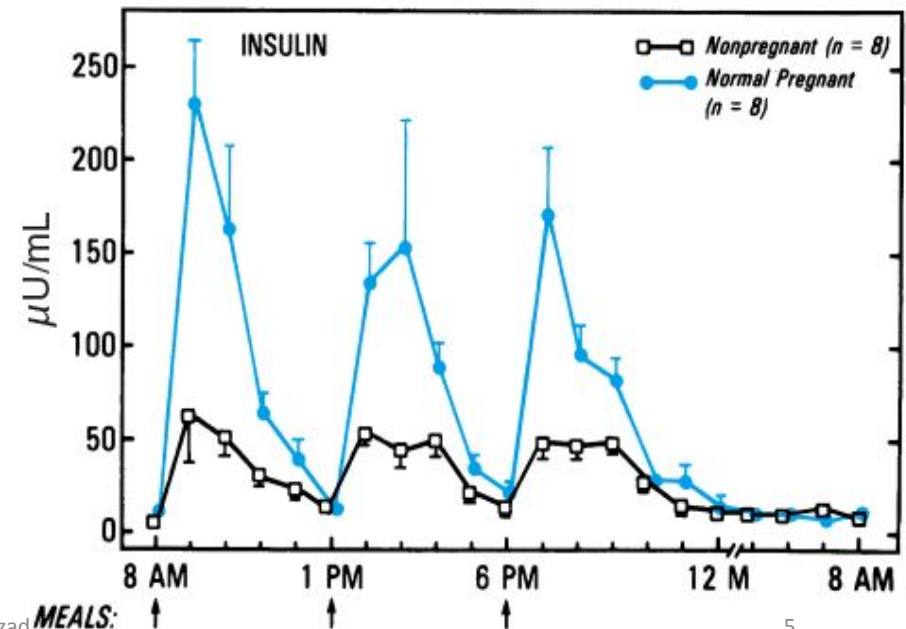
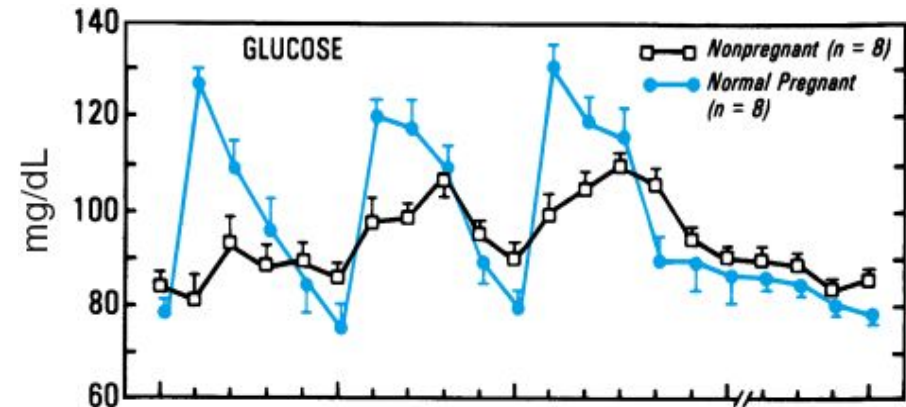
The incidence of IDD in pregnancy has been reported as approximately 1/1000 pregnancy

Less than 10% of pregnant mothers will have an indication for a GTT & of these, 20-25% will be abnormal

The incidence of gestational diabetes is reported as 10-15%, varies More in urban than rural

Pathophysiology

- ◆ Normal pregnancy is characterized by:
 - Mild fasting hypoglycemia
 - Postprandial hyperglycemia
 - Hyperinsulinemia
- ◆ Due to peripheral insulin resistance which ensures an adequate supply of glucose for the baby.



Pathophysiology

◆ Human Placental Lactogen (HPL)

- Produced by syncytiotrophoblasts of placenta.
- Acts to promote lipolysis □ increased FFA and to decrease maternal glucose uptake and gluconeogenesis. “Anti-insulin”

◆ Estrogen and Progesterone

- Interfere with insulin-glucose relationship.

◆ Insulinase

- Placental product that may play a minor role.

Carbohydrate metabolism during pregnancy

The most important reason that pregnancy uncovers the diabetic tendencies of asymptomatic women is the progressive increase in insulin resistance that occurs during gestation. Other reasons, are the increased lipolysis & the alteration in gluconeogenesis.

Insulin resistance

- 1-production of placental somatmamotropin
- 2-increased production of cortisol, estriol, & progesterone
- 3- increased insulin destruction by kidney & placenta

Risk factors:

- ◆ Diabetes mellitus (DM) type II in a first-degree relative
- ◆ Maternal obesity BMI >30
- ◆ Previous gestational diabetes
 - * previous polyhydramnios
 - * previous congenital malformation
 - * previous unexplained fetal death
 - * previous large neonate(>4000-4500)gm
 - * advanced maternal age(>35 Y)

Gestational diabetes

- ◆ GDM complicates 10–15% of pregnancies depending on the diagnostic criteria used.. Screening for diabetes in pregnancy is designed to detect previously undiagnosed type 2 diabetes and diabetes developing during pregnancy.
- ◆ Women who develop GDM are at increased risk of type 2 diabetes in later life, and education about diet and lifestyle during pregnancy can have important implications for future health.

GDM

- ◆ National Institute for Health and Care Excellence (NICE) guidelines (2015) recommend a diagnosis of GDM with a fasting glucose ≥ 5.6 mmol/l and/or a 2 hour (post-75 g glucose load) of 7.8 mmol/l.
- ◆ The WHO guidelines (2013) recommend a diagnosis with a fasting glucose of 5.1 mmol/l and/or a 1 hour (post 75 g glucose load) of 10.0 mmol/l or 2 hour of 8.5 mmol/l.

GDM

- ◆ The principles of management during pregnancy are the same as for women with pre-existing diabetes.
- ◆ Fasting $<5.3\text{mmol/l}$...1 hr postprandial $<7.8\text{mmol/l}$

Screening for diabetes in pregnancy

All antenatal women(risk factors) are screened for DM at 24-28 weeks gestation with 1 hour random oral glucose challenge test (50 gm glucose) the threshold for a positive test is $\geq$ or more than 140mg/dl.

Diagnosis of GDM is made if any two values are met or exceeded

- ◆ 2hr OGTT(done an early morning with 8-14 hr) fasting at least 3 days of unrestricted diet
- ◆ Screening test for diagnoses diabetes in people with risk factors or fasting more than 110mg/dl.
- ◆ **Universal screening not used.**

Diagnosis of GDM by OGTT

Time	100 g Glucose (American diabetes association criteria)		75 g glucose (WHO criteria)	
Fasting	95 mg/dl	5.3 mmol/L	95 mg/dl	5.3mmol/L
1-h	180mg/dl	10mmol/L	180mg/dl	10mmol/L
2-h	155mg/dl	8.6mmol/L	155mg/dl	8.6mmol/L
3-h	140mg/dl	7.8mmol/L	-	-

Investigation of cases of DM during pregnancy

- ◆ Routine antenatal urine testing for glycosuria, ketonuria & proteinuria by dipstick
- ◆ Urine analysis
- ◆ Urine culture & sensitivity if pus or bacteria are positive from urine analysis
- ◆ Renal function test

◆ Ultrasonography

- * dating scan before 12 weeks
- * at 20 weeks for fetal congenital anomalies
- * In the 3rd TM for assessment of fetal growth
- ◆ In cases of suspected IUGR, macrosomia or uncontrolled DM repeat ultrasound every 2-3 weeks

- ◆ Doppler velocimetry studies are of value in cases of IUGR
- ◆ Cardiotocograph or biophysical profile, may be used from 36 weeks gestation to assess fetal well-being.

Complication of diabetes in pregnancy

Maternal complications

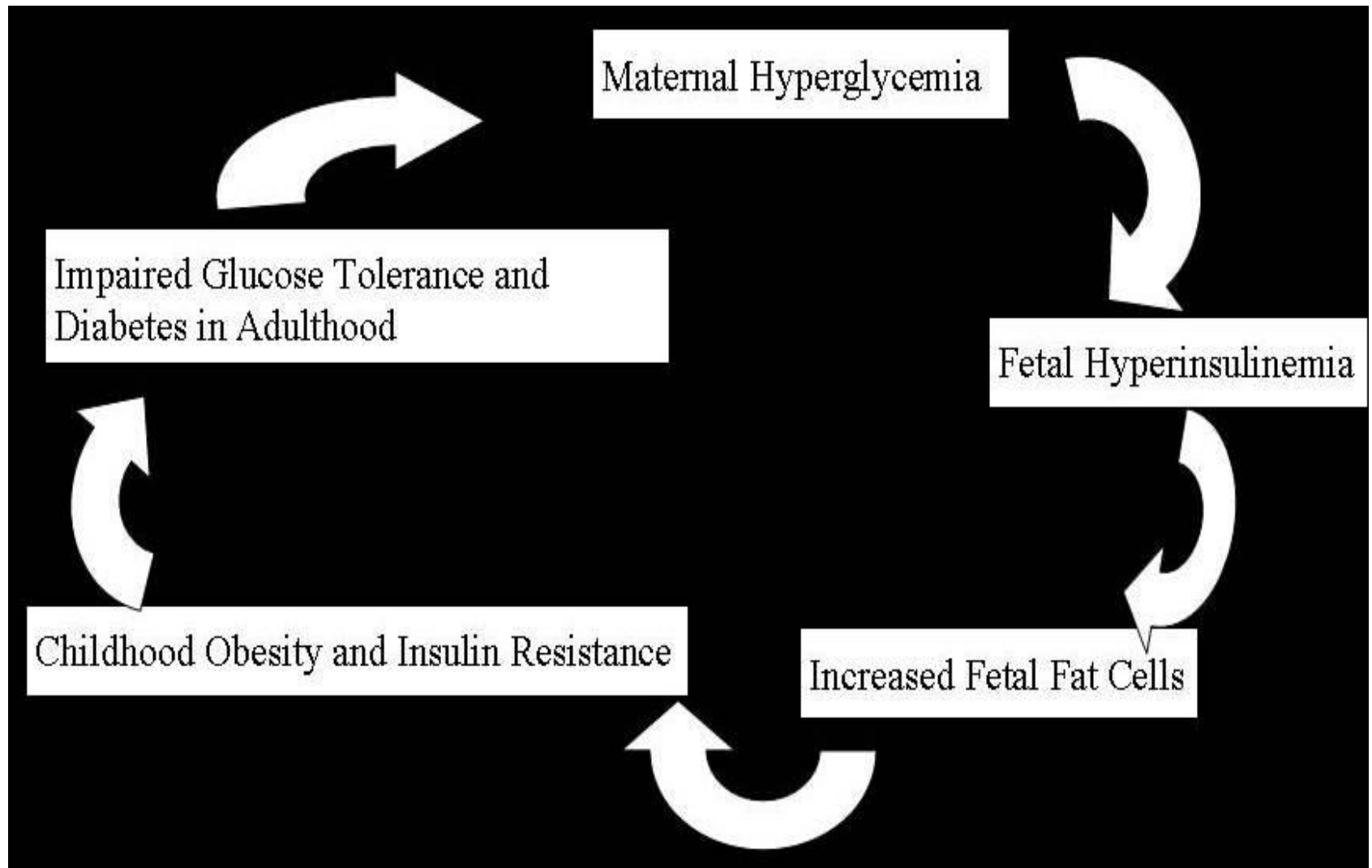
1. Preterm labour
2. Incidence of PIH may be increase
3. PPH
4. infection
5. High incidence of chorioamnionitis postpartum endometritis,UTI,candediasis of vulva & vagina

Fetal complications

- ◆ Approximate three fold increase in fetal malformation
- ◆ Increased rate of spontaneous miscarriage in poorly controlled diabetes with pregnancy
- ◆ Unexplained IUFD
- ◆ IUGR
- ◆ Fetal thrombosis

Neonatal complications

1. Respiratory distress syndrome.(RDS)
2. Neonatal hypoglycaemia
3. Polycythaemia & jaundice
4. Hypocalcaemia& hypomagnesaemia
5. Hypertrophic cardiomyopathy in infants of type I diabetes mothers.



Effects of pregnancy on diabetes

- 1- more insulin is necessary to achieve metabolic control
- 2- Progression of diabetic retinopathy
- 3- Worsening of diabetic nephropathy
- 4- Increased risk of death for patient with diabetic cardiomyopathy
- 5- ketosis
- 6- Nausea and vomiting, particularly in early pregnancy

Effects of diabetes on pregnancy

- ◆ Increased risk of miscarriage.
- ◆ Risk of congenital malformation.
- ◆ Risk of macrosomia.
- ◆ Increased risk of pre-eclampsia.
- ◆ Increased risk of stillbirth.
- ◆ Increased risk of infection.
- ◆ Increased operative delivery rate.

Factors associated with poor pregnancy outcome in diabetes

- ◆ Maternal social deprivation.
- ◆ No folic acid intake prepregnancy.
- ◆ Suboptimal approach of the woman to managing her diabetes.
- ◆ Suboptimal preconception care.
- ◆ Suboptimal glycaemic control at any stage.
- ◆ Suboptimal maternity care during pregnancy.
- ◆ Suboptimal fetal surveillance of big babies

Diabetes management

It includes both medical & obstetrical management

Medical management

Dietary recommendations:

Total caloric intake should be 30 calories/ kg for normal weight, 24 cal/ kg for obese patients

50-55% CHO

25% protein

20 % fat

Divide meals into six portions daily ,three meals & three snacks between meals. the most important snack is the bedtime snack.

Conduct an ophthalmoscopic examination beginning at 1st antenatal visit & repeated in every TM & when required

Most women with impaired glucose & GDM are initially managed with dietary advice,

- ◆ Insulin should be added to patients who have fasting hyperglycemia despite their dietary regimen
- ◆ Patients require insulin injection at least twice daily.

- ◆ A mixture of short acting and medium acting insulin is given 20-30 minutes before breakfast and dinner the dose is determined based on body weight, TM of pregnancy & sugar level
- ◆ The initial dose is
 - ◆ 1st TM 0.6 units /kg
 - ◆ Second TM 0.7 u/kg
 - ◆ 3rd TM 0.8 u /kg

Obese patient usually require 0.1 u above the recommended dose in each TM

Total daily dose is calculated & $\frac{2}{3}$ of the dose is given in the morning & the remaining $\frac{1}{3}$ is given in the evening

the morning dose is divided as $\frac{2}{3}$ NPH & $\frac{1}{3}$ regular insulin ,while the evening dose is divided as $\frac{1}{2}$ regular & $\frac{1}{2}$ NPH

insulin is given by SC route with an insulin syringe in the thigh ,upper arm ,buttocks or abdominal wall.

Types of insulin

- ◆ 1-Rapid acting insulin action 15min peak in 1 to 2 hr last between 2-4hr(lispro)
- ◆ 2-Regular or short acting ...30min..peak 2-3hr, last 3-6hr
- ◆ 3-intermediate actingact after 2-4hr,peak 4-12hr and last for 12-18hr
- ◆ 4-long acting reach blood stream after several hr and lower blood glucose level up to 24hr.
- ◆ 5-ultra –long..start after 6hr and works upto36hrs.

◆ The dose is readjusted according to the fasting & pre-prandial blood sugar levels

She should have 5 times per day checking her levels

Of blood sugar fasting ,3 times postprandial 1 hour and bedtime.if the case receive insulin .

Hypoglycemia

Warning signs of hypoglycemia vary from person to person include

Feeling shaky, sweating, tingling in the lips, going pale ,heart pounding, confusion & irritability.

Management of hypoglycemia

- ◆ Ensure that those close to the patients have been instructed in the 1st recognition & 1st aid managements of hypoglycemia
- ◆ 1st aid management consists of administration of a sugary fluid followed by a starchy snack
- ◆ If left untreated & the patient becomes unconscious administration of IV glucose 10-25% in a central vein is indicated.
- ◆ Glucagon injections are preferable if available.

Obstetrical management

Preconception care:

Women who are known to be diabetic prior to pregnancy should have controlled blood sugar level , glycosylated Hb gives an idea about diabetic control over the three months

Normal HbA1c up to $<6.5\text{mg/dl}$

High-dose folic acid (5 mg daily) to reduce the risk of neural tube defects

Planning periconception adjustments to other medications such as statins and angiotensin-converting enzyme (ACE) inhibitors before pregnancy

- Poor glycaemic control is associated with a significantly increased risk of congenital anomalies, particularly neural tube defects and cardiac anomalies.
- The most critical period for the embryo is therefore the period of organogenesis, which occurs in the first 42 days of pregnancy, and this is often before the pregnancy is medically confirmed.
- The level of HbA1c in early pregnancy also correlates with the risk of early fetal loss.

Antenatal care

- 1-Strict control of plasma glucose around the time of conception will prevent congenital anomalies, PET, polyhydramnios, preterm labour & fetal macrosomia.
- 2- early booking & dating of the pregnancy.
Dating is confirmed by two measurements of the BPD between 14-24 weeks
One measurement is usually made around 16 weeks when major congenital malformations may be diagnosed , serum alpha fetoprotein estimation may be performed at the same time

GDM

- The American College of Obstetricians and Gynecologists typically recommends the following ranges for blood glucose levels:
- Before a meal: 95 milligrams (mg)/dL or less. (Mg/dL stands for milligrams per decilitre.)
- One hour after a meal: 140 mg/dL or less.
- Two hours after a meal: 120 mg/dL or less.

3- as pregnancy progresses ANC should be provided at 2 weekly interval, with a view to early detection of maternal complications & assessing fetal welfare .after 24 weeks estimate of fetal growth are performed at least monthly

4-antenatal CTG & kick count

5- in late pregnancy admission to hospital at 36 weeks has been the practice until recently , but if control is good & no complications have arisen, admission to hospital may be further delayed.

6-limit weight gain throughout gestation to 9-14 kg(20-30lb)

Delivery

Timing

If pregnancy is apparently normal it may be allowed to continue up to 40 weeks.

In more complicated cases delivery is usually at 38 weeks, & occasionally elective preterm delivery become necessary most often because of PET

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Mode of delivery

C.S may be indicated if when complications dictate urgent delivery& when malpresentation is present.

C.S is not indicated on account of diabetes alone .

Active labour

- ◆ Monitor blood glucose & insulin infusions frequently (every $\frac{1}{2}$ -1 hour) to keep blood glucose level between (80-110 mg/d
 - ◆ If induction of labour is planned
 - ◆ No insulin is given on the morning induction & no breakfast
 - ◆ Induction is started early in the morning
 - ◆ Check a capillary glucose level with the glucometer
 - ◆ Insulin is added to the infused fluid & is not given SC
- The same regimen is employed when delivering by elective CS

Postpartum care

- *encourage breast feeding
- *women with pre-existing diabetes should revert to their pre-pregnancy insulin regimen, those with GDM who were treated with insulin usually do not require any treatment
- *Increase dietary carbohydrate by 50 g & adjust insulin dosage
- *discuss family planning methods
- *six weeks after delivery a 75gm OGTT should be carried out.or HBA1C



◆ THANK YOU